

Reinforcement Learning for Recommendation Systems: A Survey

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Abstract

Reinforcement learning has emerged as a powerful paradigm for recommendation systems, as modern recommenders increasingly operate as sequential decision-making processes rather than static prediction models. Traditional supervised learning approaches are effective at optimizing short-term metrics such as click-through rate, but struggle to capture delayed feedback, user preference dynamics, and long-term value. Reinforcement learning (RL), with its ability to model interactions over time and optimize cumulative rewards, provides a principled framework for addressing these challenges.

In this survey, we provide a systematic review of reinforcement learning-based recommendation systems (RLRS). We first introduce the background and problem formulation of applying RL to recommendation scenarios. We then present a structured taxonomy that categorizes existing methods from the perspectives of learning paradigms, state representation, and exploration strategies. Representative works are discussed to highlight key technical innovations and practical insights. Furthermore, we analyze emerging research trends, including deep exploration and the integration of large language models (LLMs) with RL-based recommenders. Finally, we summarize open challenges and future research directions, aiming to provide a comprehensive reference for both academic researchers and industrial practitioners.

1. Introduction

Reinforcement learning has become an increasingly important approach for recommendation systems, as real-world recommendation tasks are inherently sequential and involve long-term interactions between users and platforms. Early recommendation systems were primarily formulated as static prediction problems, where models estimate the relevance of an item to a user based on historical interactions. While supervised learning approaches have achieved notable success in optimizing short-term engagement metrics, they are fundamentally limited in modeling delayed feedback and long-term user satisfaction.

In real-world scenarios, recommendation is inherently a sequential decision-making process. Each recommendation influences future user behavior, preference evolution, and engagement patterns. Optimizing only short-term metrics may lead to suboptimal outcomes such as user fatigue, reduced diversity, or long-term churn. This gap between short-term optimization and long-term value has motivated increasing attention toward reinforcement learning (RL), which naturally models interaction as a Markov Decision Process (MDP) and optimizes cumulative rewards over extended horizons [9].

The application of reinforcement learning to recommendation systems has attracted growing interest over the past decade. Early studies explored value-based and policy-based methods to address large action spaces and delayed rewards, including Deep Q-Networks (DQN) and actor-critic frameworks [3, 12]. Subsequent research focused on improving state representation, exploration efficiency, and scalability, enabling RL-based methods to better align with industrial-scale recommendation settings [11]. More recently, the community has shifted toward deeper forms of exploration and long-term value optimization, supported by high-fidelity simulators and system-level designs [4].

In parallel, the rapid advancement of large language models (LLMs) has introduced new possibilities for recommendation systems. By incorporating reasoning, planning, and tool usage capabilities, LLM-powered agents have begun to complement reinforcement learning in sequential recommendation tasks, leading to hybrid architectures that extend beyond traditional model-centric designs [10]. These developments signal a paradigm shift from isolated algorithmic improvements toward holistic, agent-based recommendation systems.

This survey aims to provide a comprehensive and structured overview of reinforcement learning-based recommendation systems. Specifically, we make the following contributions:

- We formalize recommendation systems from a reinforcement learning perspective and summarize common problem formulations.
- We propose a taxonomy that categorizes RL-based recommendation methods according to learning paradigms, state representation strategies, and exploration mechanisms.
- We review representative works to highlight key technical innovations and practical lessons.
- We analyze emerging trends, including deep exploration and the integration of RL with large language models, and discuss open challenges and future directions.

By synthesizing existing research through a unified lens, this survey seeks to facilitate a deeper understanding of how reinforcement learning can address the core challenges of modern recommendation systems.

2. Background and Problem Formulation

In this section, we introduce the fundamental background of reinforcement learning and formalize recommendation systems from a sequential decision-making perspective. We also discuss commonly used evaluation protocols and simulation environments that support offline training and analysis of reinforcement learning-based recommendation systems.

2.1. Reinforcement Learning Preliminaries

Reinforcement learning (RL) studies how an agent interacts with an environment to maximize cumulative rewards through sequential decision making [9]. A standard RL problem is commonly formulated as a Markov Decision Process (MDP), defined by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$, where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, $\mathcal{P}(s'|s, a)$ the state transition probability, $\mathcal{R}(s, a)$ the reward function, and $\gamma \in (0, 1]$ the discount factor.

At each time step t , the agent observes a state $s_t \in \mathcal{S}$, selects an action $a_t \in \mathcal{A}$ according to a policy $\pi(a|s)$, receives a scalar reward r_t , and transitions to the next state s_{t+1} . The objective of reinforcement learning is to learn an optimal policy that maximizes the expected discounted return:

$$\mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]. \quad (1)$$

Modern RL algorithms can be broadly categorized into value-based methods, policy-based methods, and actor-critic methods [6, 7]. These approaches form the algorithmic foundation of reinforcement learning-based recommendation systems.

2.2. MDP Formulation of Recommendation Systems

Recommendation systems can be naturally modeled as sequential decision problems, where the system repeatedly interacts with users and influences their future behaviors [8]. In an RL-based recommender system, each user interaction corresponds to a step in the MDP.

2.2.1. State Space

The state s_t is designed to capture the current context of the user and the system. Typical state representations include:

- User profiles and static attributes;
- Historical interaction sequences (clicked, viewed, or purchased items);
- Contextual information such as time, device, or location.

Early approaches relied on raw or handcrafted features, while more recent methods adopt dense vector representations learned via deep neural networks [11].

2.2.2. Action Space

The action space corresponds to the recommendation decisions made by the system. Depending on the task, an action may represent:

- Recommending a single item;
- Recommending a ranked list (slate) of items;
- Selecting a composite action such as item-category or item-layout pairs.

Slate recommendation significantly enlarges the action space and introduces combinatorial complexity, posing major challenges for conventional RL algorithms [5].

2.2.3. Reward Design

The reward function encodes the optimization objective of the recommendation system. Commonly used reward signals include clicks, dwell time, purchases, and ratings. However, these signals are often sparse, delayed, and noisy. Moreover, optimizing immediate rewards does not necessarily align with long-term user satisfaction or platform objectives [4].

Designing reward functions that effectively reflect long-term value remains one of the central challenges in reinforcement learning-based recommendation systems.

2.3. Evaluation Protocols and Simulation Environments

Evaluating RL-based recommendation systems is non-trivial due to the interactive nature of the problem. Online A/B testing provides the most reliable evaluation but is expensive and risky. As a result, offline evaluation and simulation-based methods are widely adopted.

2.3.1. Offline Evaluation

Offline evaluation typically relies on logged historical data and off-policy evaluation techniques. Methods such as inverse propensity scoring and doubly robust estimators have been explored to correct for distribution shift between the behavior policy and the learned policy [3]. However, offline evaluation remains fundamentally limited by the quality and coverage of logged data.

2.3.2. Simulation-Based Evaluation

Simulation environments provide a controllable and scalable alternative for evaluating RL-based recommendation algorithms. User behavior simulators attempt to model user responses to recommendation sequences, enabling long-horizon evaluation and deep exploration [4]. High-fidelity simulators have been shown to play a critical role in advancing long-term value optimization in industrial recommendation systems.

Overall, background modeling choices, problem formulation, and evaluation protocols jointly determine the effectiveness and applicability of reinforcement learning methods in recommendation systems.

3. A Taxonomy of Reinforcement Learning-based Recommendation Systems

Existing reinforcement learning-based recommendation systems (RLRS) differ significantly in how they formulate the learning objective, represent user states, and handle exploration under large action spaces. In this section, we present a concise taxonomy that categorizes prior work along three orthogonal dimensions: learning paradigms, state representation strategies, and exploration mechanisms.

3.1. Learning Paradigms

From an algorithmic perspective, RL-based recommendation methods can be broadly categorized according to their underlying learning paradigms.

3.1.1. Value-based Methods

Value-based approaches estimate the expected return of state–action pairs and derive policies by maximizing action values. Early RLRS works adopted Deep Q-Networks (DQN) to model recommendation as a sequential decision problem [7]. While effective for small action spaces, vanilla DQN struggles with large-scale recommendation scenarios due to the combinatorial explosion of candidate items.

To address this issue, subsequent studies introduced structured action decompositions and approximation techniques, such as slate-based value decomposition, enabling value-based learning in list-wise recommendation settings [5].

3.1.2. Policy-based and Actor–Critic Methods

Policy-based methods directly optimize parameterized policies and are better suited for large or continuous action spaces. Actor–critic architectures, which combine policy learning with value estimation, have been widely adopted in industrial recommendation systems due to their stability and scalability [6, 11].

These methods allow flexible policy parameterization and naturally support stochastic policies, making them particularly suitable for long-term recommendation optimization.

3.2. State Representation Strategies

State representation plays a central role in RLRS, as it determines how user preferences and interaction histories are encoded.

Early approaches relied on raw or handcrafted features, such as recent clicks or item statistics. More advanced methods model user behavior sequences using recurrent neural networks or attention mechanisms, producing dense low-dimensional embeddings that summarize long-term interests [12].

Recent work increasingly adopts representation learning techniques that jointly optimize state embeddings and recommendation policies, enabling end-to-end training and improved generalization across users and contexts.

3.3. Exploration Strategies

Efficient exploration is critical in recommendation systems, where user feedback is sparse and interactions are costly.

Traditional exploration mechanisms, such as ϵ -greedy or entropy regularization, provide limited long-term coverage of the recommendation space. To overcome these limitations, recent studies advocate *deep exploration*, which explicitly considers the long-term impact of exploratory actions on future user states [4].

By leveraging high-fidelity simulators and long-horizon evaluation, deep exploration methods have demonstrated substantial improvements in long-term value metrics, highlighting the importance of exploration-aware system design.

3.4. Summary

In summary, reinforcement learning-based recommendation systems can be systematically understood through the lenses of learning paradigms, state representation, and exploration strategies. While early research focused on adapting standard RL algorithms to recommendation tasks, recent advances increasingly emphasize system-level integration and long-term value optimization. This taxonomy provides a structured foundation for analyzing representative methods and emerging trends, which we discuss in the following sections.

4. Representative Methods and Case Studies

In this section we discuss several representative works that capture the major technical directions and practical lessons in reinforcement learning based recommendation systems. We focus on three complementary perspectives: (1) broad surveys and frameworks that synthesize the field, (2) methodological advances addressing sparse/delayed feedback via deep exploration, and (3) recent, large-scale simulator-based RL systems for web-scale recommendations.

4.1. Cornerstone Survey: Afsar et al. (2021)

Afsar et al. (2021) provide a comprehensive survey of reinforcement learning based recommender systems (RLRS), organizing the field into four components: state representation, policy optimization, reward formulation, and environment construction. The paper synthesizes early DRL adaptations for recommendation, catalogs evaluation methodologies, and highlights practical challenges such as large action spaces and offline evaluation bias. As a field-level reference it is especially useful for researchers who want a systematic entry point into RLRS. [1] :contentReference[oaicite:1]index=1

4.2. Deep Exploration for Sparse and Delayed Feedback

One fundamental challenge for RL in recommendation is sparse and delayed rewards: typical short interactions produce sparse signals that make naive exploration ineffective. Zhu & Van Roy (2021) develop the notion of *deep exploration* tailored to recommendation settings, proposing algorithms and uncertainty-aware architectures that prioritize exploration actions with long-term informational value. They demonstrate through high-fidelity simulators that deep exploration strategies can substantially outperform single-step exploration baselines on long-horizon metrics. This work is influential in reframing exploration from a myopic (single-step) notion to a planning-oriented, long-horizon objective. [13] :contentReference[oaicite:2]index=2

4.3. RecoMind: Simulator-based, Web-scale RL for Session Satisfaction

RecoMind (Ayed et al., 2025) presents a practical, simulator-based RL framework designed for web-scale session optimization. RecoMind integrates existing supervised recommenders to bootstrap a simulation environment, applies custom exploration strategies suited for extremely large item catalogs, and demonstrates both offline simulation gains and positive online A/B test results (notably improvements in consumption and session depth). The system highlights engineering patterns for deploying RL at scale: simulator bootstrapping, policy warm-starting, and tailored exploration for massive action spaces. RecoMind is an important example of how research ideas (deep exploration, simulator-based evaluation) can be integrated into production-like pipelines. [2] :contentReference[oaicite:3]index=3

4.4. Cross-cutting Observations

- **Survey → Practice:** The field has matured from algorithmic adaptations (DQN, actor–critic) to frameworks and system designs that address evaluation, simulators, and deployment—exactly the components emphasized in Afsar et al. [1].
- **Exploration matters:** Advances in exploration (e.g., epistemic uncertainty, deep exploration) are not merely theoretical: they drive practical gains when paired with good simulators and long-horizon evaluation, as shown by both Zhu & Van Roy and recent system papers. [2, 13]
- **Engineering patterns:** Successful web-scale RL deployments often combine (i) simulator-based offline training, (ii) warm-starting from supervised models, and (iii) exploration strategies tuned to large catalogs—patterns clearly demonstrated in RecoMind. [2]

These representative works together illustrate the trajectory of RL-based recommendation: from conceptual surveys to methodological advances (especially in exploration) and finally to system-level integration needed for real-world, web-scale adoption.

5. Emerging Trends and Research Directions

Recent advances in reinforcement learning based recommendation systems indicate a clear shift from isolated algorithmic improvements toward holistic system-level designs and long-term value optimization. In this section, we summarize several emerging trends that are reshaping the research landscape of RL-based recommendation.

5.1. From Short-term Metrics to Long-term Value

A prominent trend in modern recommendation research is the transition from optimizing short-term metrics, such as click-through rate or immediate engagement, to optimizing long-term user value. Reinforcement learning naturally supports this paradigm by modeling recommendation as a sequential decision-making problem with delayed rewards.

Recent works demonstrate that policies optimized for long-horizon objectives can significantly outperform myopic baselines when evaluated on user retention, session length, and cumulative satisfaction [2, 13]. This shift also motivates the development of better simulators and evaluation protocols capable of reflecting long-term user dynamics.

5.2. System-level Integration and Simulator-based Training

As RL-based recommendation systems mature, research focus has expanded from standalone models to system-level integration. Modern systems increasingly rely on simulators to enable scalable offline training, safe exploration, and long-horizon evaluation. Simulator-based pipelines allow reinforcement learning agents to explore diverse recommendation strategies without incurring real-world risks, serving as a critical bridge between offline research and online deployment [1, 2].

This trend highlights that effective RLRS design requires not only algorithmic advances, but also careful engineering of data pipelines, simulation fidelity, and deployment constraints.

5.3. Integration of Reinforcement Learning and Large Language Models

The rapid progress of large language models (LLMs) has introduced new opportunities for reinforcement learning-based recommendation systems. By incorporating reasoning, planning, and tool-use capabilities, LLMs can complement RL agents in handling complex decision contexts, sparse feedback, and high-level user intent modeling.

Recent studies explore hybrid architectures where RL optimizes long-term objectives while LLMs assist with state abstraction, action generation, or policy initialization. Such integrations suggest a shift from purely model-centric approaches toward agent-based recommendation systems that combine learning, reasoning, and interaction capabilities.

5.4. Multi-objective and Responsible Recommendation

Another emerging direction is the consideration of multiple, potentially conflicting objectives, such as accuracy, diversity, fairness, and user well-being. Reinforcement learning provides a flexible framework for incorporating multi-objective optimization through reward shaping or constrained policy learning.

At the same time, concerns about fairness, interpretability, and safety are gaining increasing attention. Future RL-based recommender systems are expected to explicitly address these issues, moving beyond performance-centric evaluation toward more responsible and transparent recommendation policies.

5.5. Summary

Overall, emerging trends in reinforcement learning based recommendation systems reflect a broader paradigm shift: from short-term optimization to long-term value, from isolated algorithms to integrated systems, and from single-objective learning to multi-faceted, responsible recommendation. These trends set the stage for the open challenges and future research directions discussed in the next section.

6. Challenges and Open Problems

Despite significant progress, reinforcement learning based recommendation systems still face several fundamental challenges that limit their broader adoption and practical impact.

6.1. Reward Design and Long-term Alignment

Designing reward functions that faithfully reflect long-term user value remains a central challenge. Observable signals such as clicks or dwell time are often sparse and biased, and may not correlate with long-term satisfaction or retention. While long-horizon optimization mitigates short-term bias, defining rewards that align business objectives with user well-being remains an open problem.

6.2. Efficient and Safe Exploration

Exploration is inherently costly in recommendation systems, as suboptimal actions directly affect user experience. Although deep exploration methods improve long-term learning efficiency, they often rely on high-quality simulators. How to balance efficient learning, safety constraints, and real-world deployment risks remains unresolved.

6.3. Scalability and Deployment Constraints

Real-world recommendation systems operate at massive scale, with millions of users and items. Many RL algorithms face practical bottlenecks in computation, latency, and system integration. Bridging the gap between algorithmic advances and production-ready systems is still a major engineering challenge.

6.4. Evaluation and Reproducibility

Reliable evaluation of RL-based recommenders is difficult due to distribution shift and feedback loops. Simulator-based evaluation alleviates some issues, but reproducibility and generalization across domains remain limited. Establishing standardized benchmarks and evaluation protocols is an important direction for future research.

7. Conclusion

This survey reviewed the development of reinforcement learning based recommendation systems from early algorithmic adaptations to modern system-level architectures. We presented a structured taxonomy, discussed representative works, and summarized emerging trends that emphasize long-term value optimization, simulator-based training, and integration with large language models.

Overall, the field is undergoing a paradigm shift from short-term, single-step optimization toward holistic, long-horizon decision making. Reinforcement learning provides a principled framework for this transition, but its success in recommendation systems depends not only on algorithmic innovation, but also on advances in system design, evaluation methodology, and responsible deployment. We hope this survey serves as a useful reference and inspires future research in this rapidly evolving area.

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